

INTRODUCTION TO NETWORK DEVICES

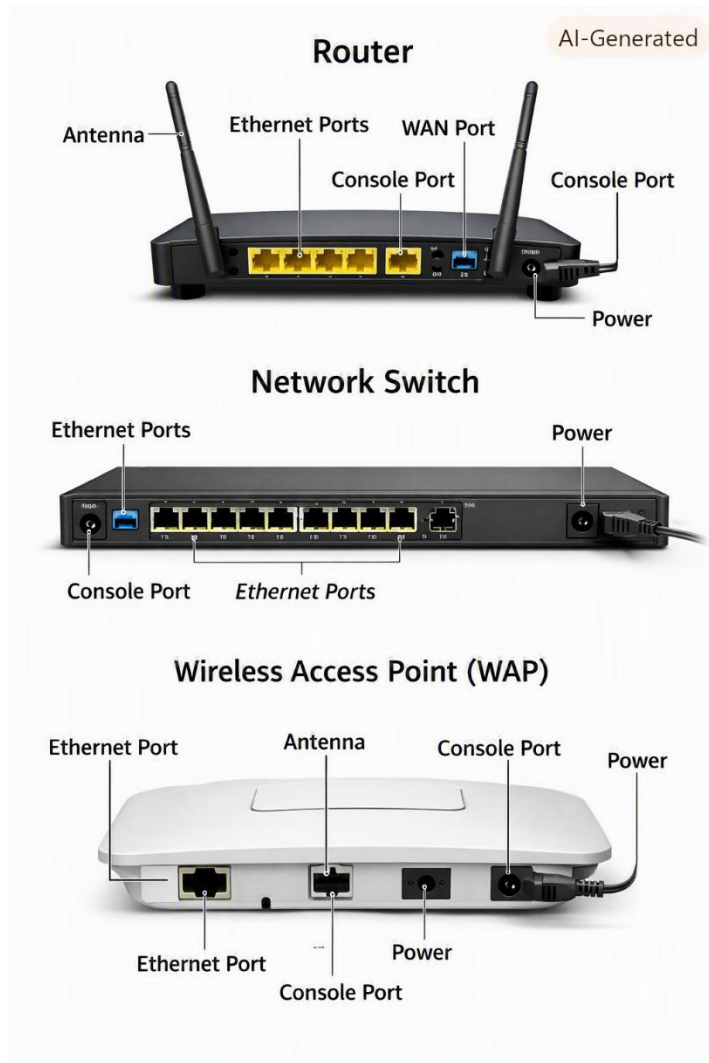
THEORY

1. Create a table listing the main functions and typical ports used by switches, routers, hubs, firewalls, and access points in a home WiFi setup.

Networking Devices Functions and Ports in a Home WiFi Setup

Device	Main Function	Typical Ports Used
Router	Connects my home network to the internet and forwards data between different networks. It also provides WiFi connectivity and assigns IP addresses using DHCP.	WAN/Internet Port, LAN Ethernet Ports (RJ45), WiFi Wireless Connection
Switch	Connects multiple wired devices in the same local network and allows devices to communicate with each other.	Multiple Ethernet (RJ45) Ports
Hub	Connects multiple devices in a network and broadcasts incoming data to all connected devices.	Multiple Ethernet (RJ45) Ports
Firewall	Protects the home network by monitoring and filtering incoming and outgoing network traffic.	WAN Port, LAN Port, Management Port
Access Point	Provides wireless network access to devices such as laptops, smartphones, and smart TVs.	Ethernet Port (RJ45) for wired connection and Wireless Radio (2.4GHz/5GHz)

2. Given images or diagrams of network devices, label each device and mark the ports (Ethernet, console, power, antenna) for at least three devices.



3. Pick a popular app you use daily (like Instagram, Zomato, or Paytm) and describe in 3-4 lines how at least two different network devices (from switches, routers, firewalls, access points) play a role when you use that app.



Role of Network Devices While Using Instagram App

I use the Instagram app daily to watch videos, upload photos, and communicate with others. When I use Instagram, my WiFi router/access point connects my mobile device to the internet and transfers data between my home network and Instagram servers. The firewall protects my network by monitoring and filtering unwanted traffic while allowing safe

communication. The switches used in data centers help connect multiple servers and manage data transfer efficiently.

4. Suppose your college WiFi is slow and sometimes disconnects. Based on the features of access points, switches, and routers, suggest one possible reason for the issue and how upgrading one device could help.

Hint: Think about device capacity, coverage, and traffic handling.



One possible reason for the slow and unstable WiFi connection is that the **wireless access point (WAP)** being used has **limited capacity and outdated technology**. When too many students connect at the same time, the access point cannot handle the high traffic efficiently, causing delays and disconnections.

Upgrading to a **dual-band or Wi-Fi 6 access point** would improve performance by providing **greater coverage, faster data transfer rates, and better traffic management**. This upgrade allows multiple devices to connect simultaneously without congestion, ensuring smoother internet access across the campus.